

L2 Norm Analysis Report

Four-Head Transformer Variant

Dataset: Four-Head Transformer on (a+b) mod 97
Training Configuration: 40,000 epochs, 30/70 train/test split
Model: 4 attention heads, d_model=128, head_dim=32
Dropout predictor: rate=0.9 (post-training stress test)

REPORT PURPOSE:

This report presents comprehensive analysis of L2 norm (weight magnitude) behavior across three independent runs of the four-head transformer.

KEY FINDINGS:

- Grokking occurs at significantly different epochs across runs (149, 2815, 671)
- L2 norm values show high variability despite identical architecture
- Dropout Gap is consistently negative across all runs (-0.97 to -0.98)
- L2 norm decay pattern differs substantially between runs
- Single-run metrics cannot be trusted for prediction without averaging

Table 1: Raw Metrics Across All Runs

Run	Total Epochs	Initial L2	Final L2	Total Decay	Decay %
Run 1	40,000	115.7222	65.2405	50.4817	43.62%
Run 2	40,000	116.4240	39.1517	77.2723	66.37%
Run 3	40,000	115.6342	38.9024	76.7318	66.36%

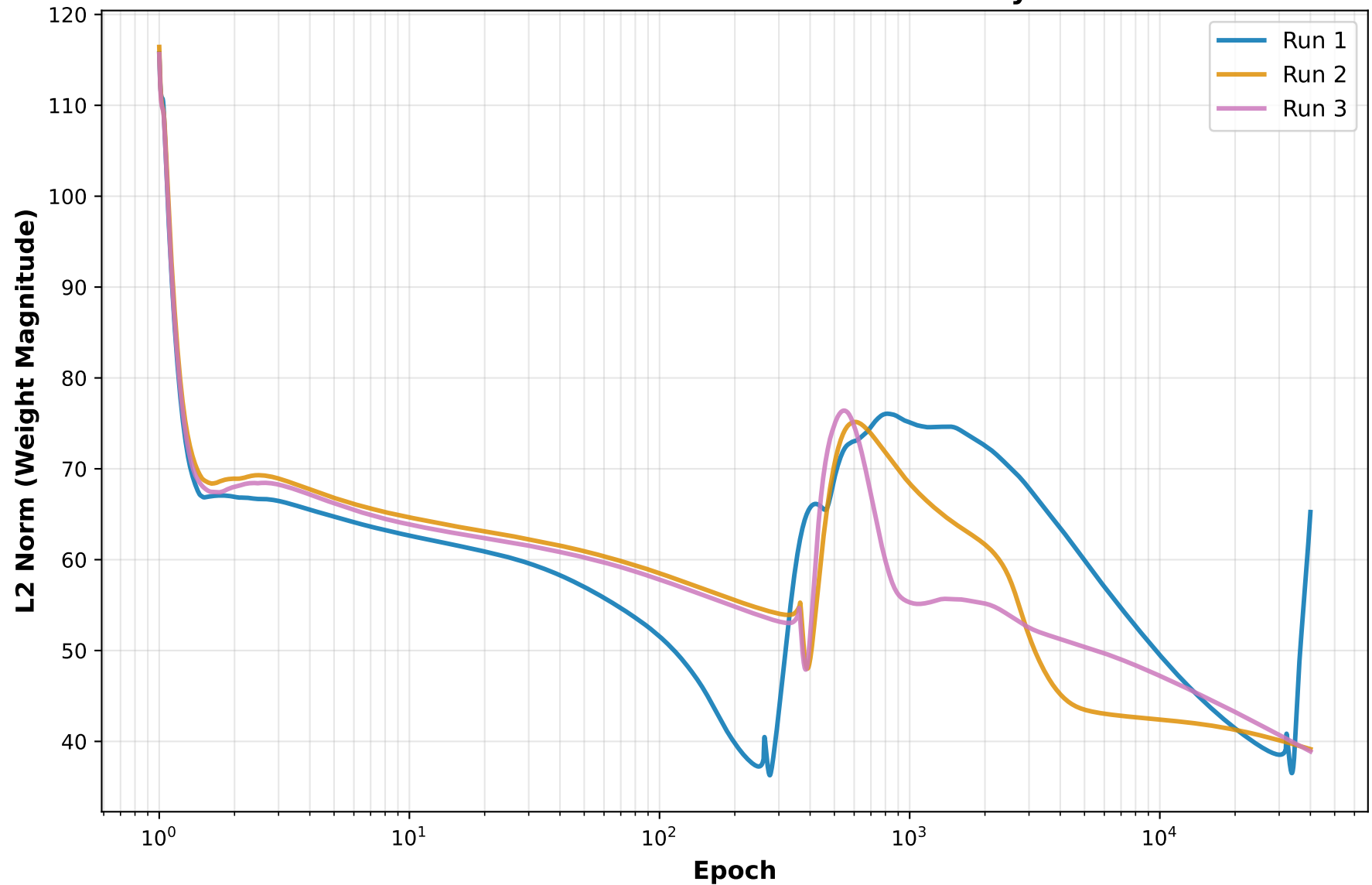
Table 2: Test Accuracy and Loss

Run	Initial Test Acc	Final Test Acc	Initial Loss	Final Loss	Grok Epoch
Run 1	0.010931	1.000000	4.897544	0.001403230	149
Run 2	0.010323	1.000000	5.265750	0.000001369	2815
Run 3	0.009109	1.000000	4.998405	0.000000873	671

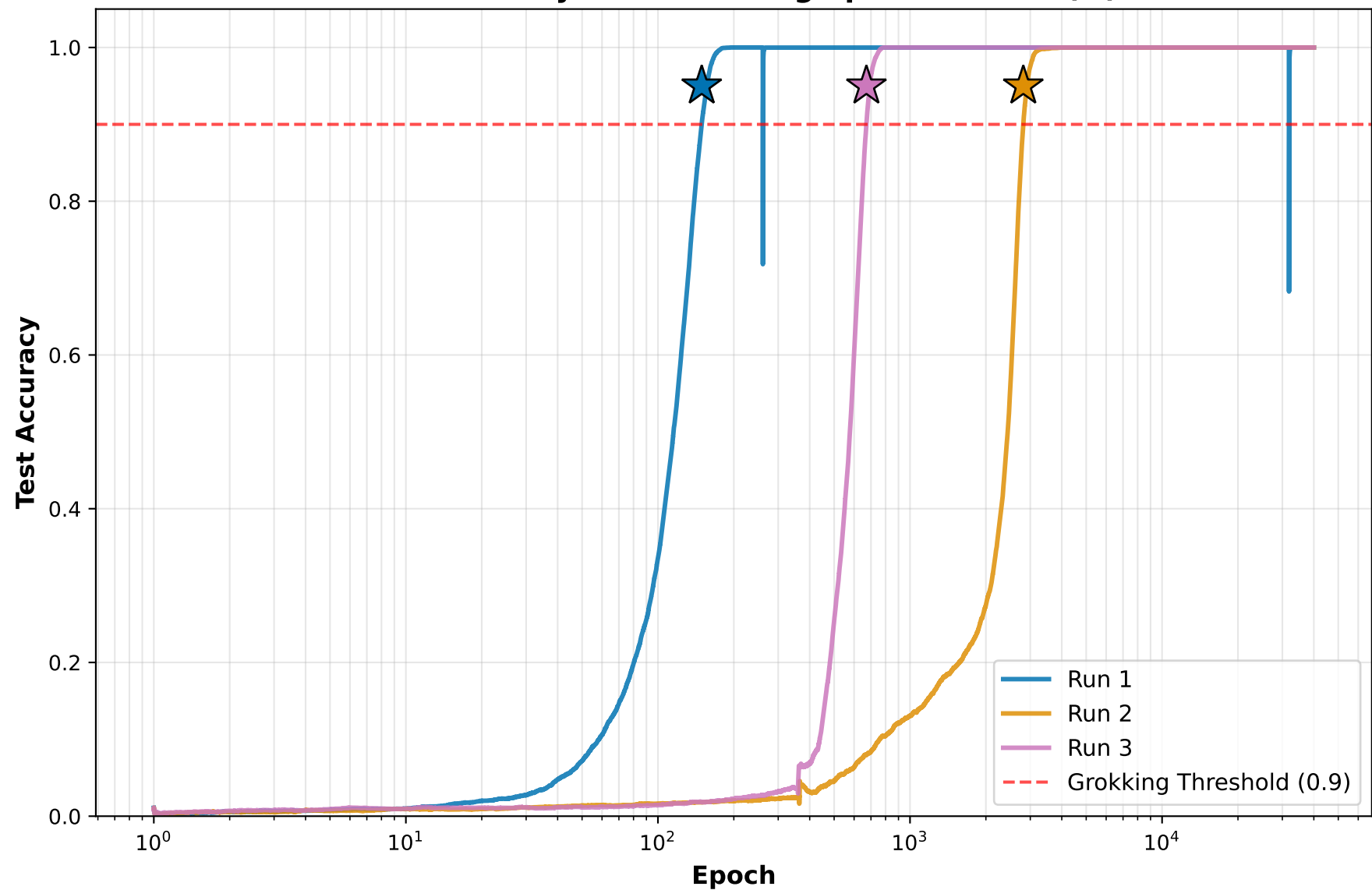
Table 3: Dropout Gap (Stress Test Results)

Run	Min Dropout Gap	Final Dropout Gap	Max Dropout Gap
Run 1	-0.988007	-0.966297	0.010627
Run 2	-0.984819	-0.976772	0.011538
Run 3	-0.981934	-0.968726	0.008957

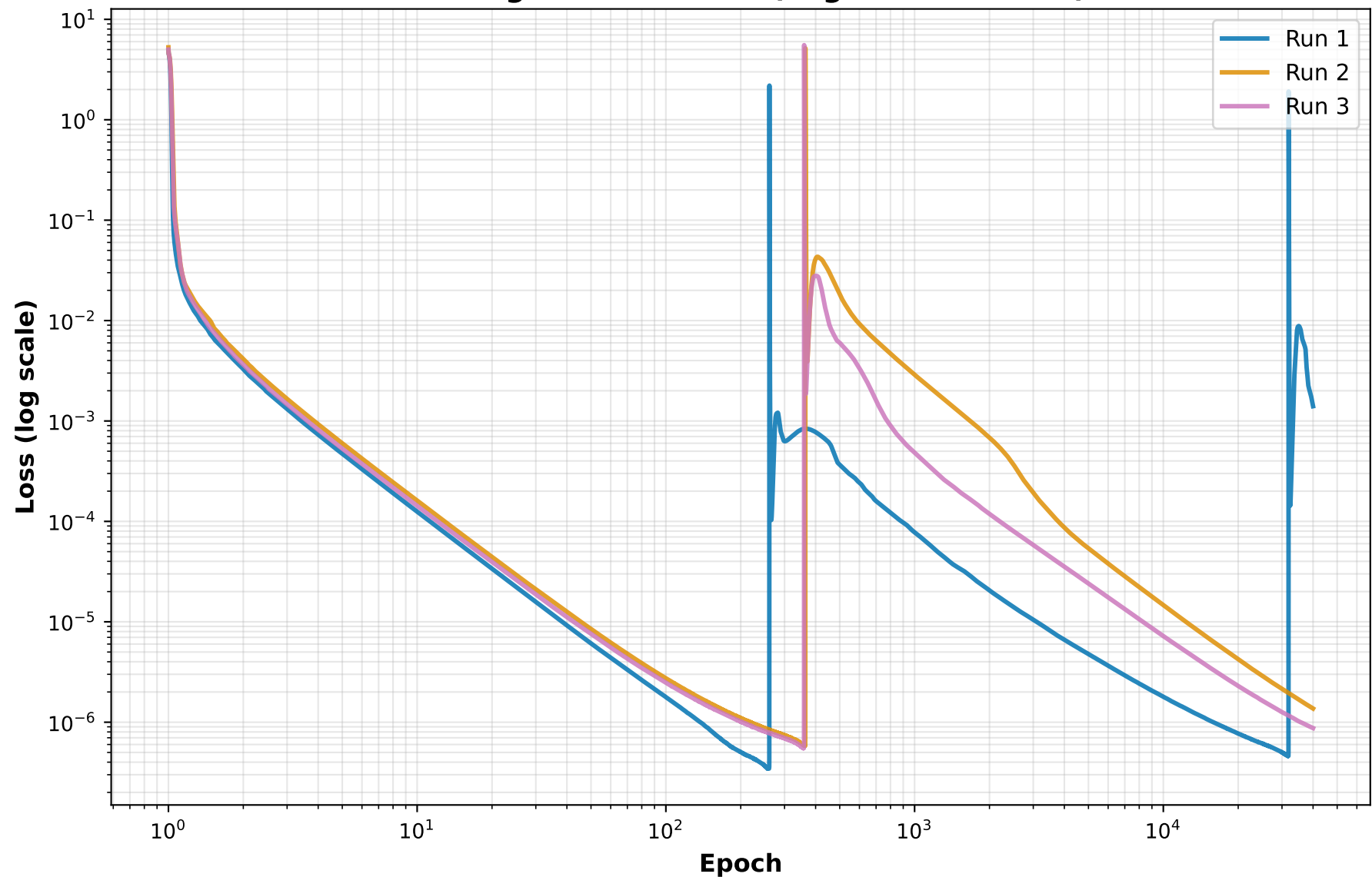
L2 Norm Evolution: All Runs Overlay



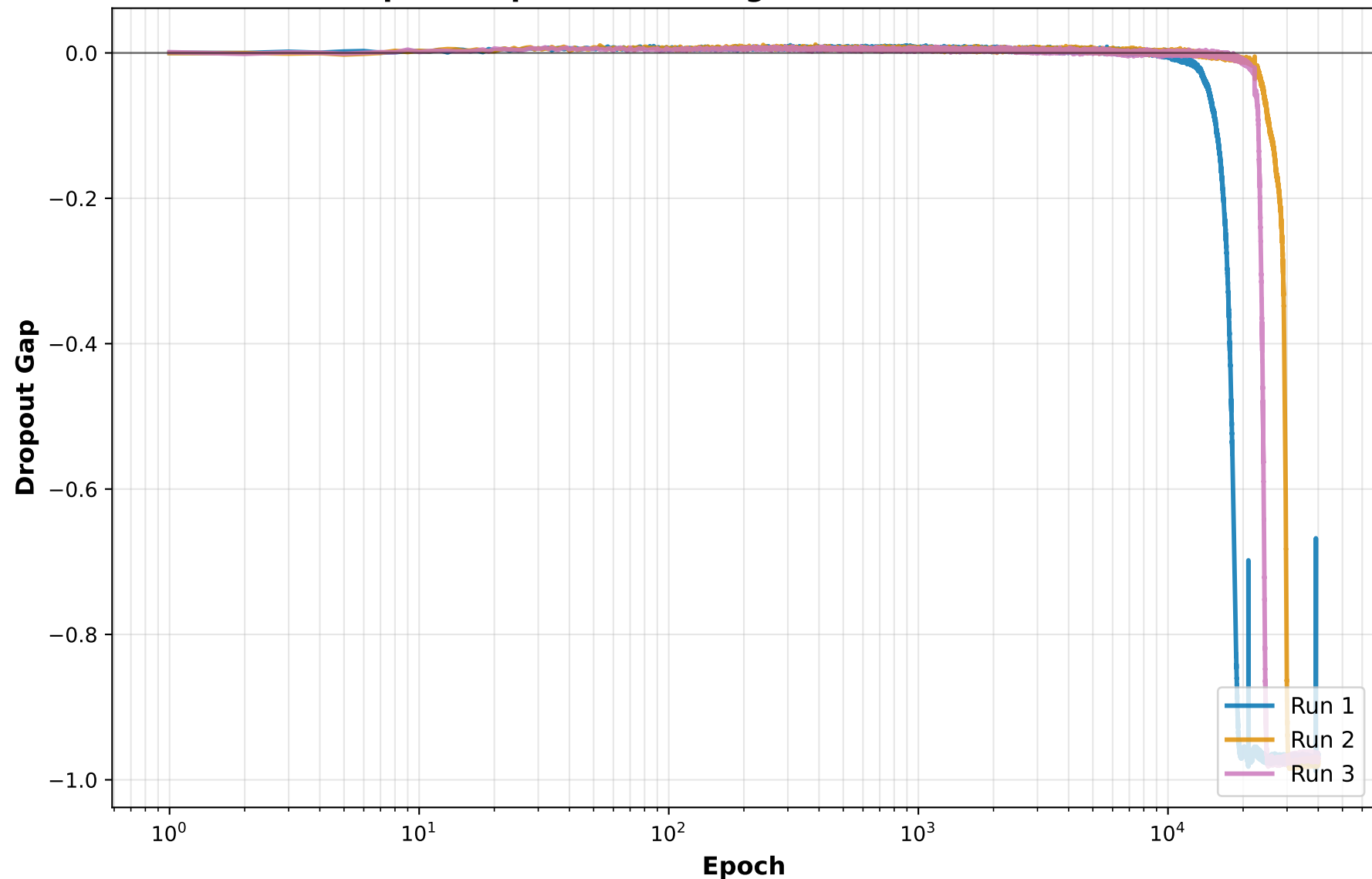
Test Accuracy with Grokking Epoch Marked (★)



Training Loss: All Runs (Logarithmic Scale)

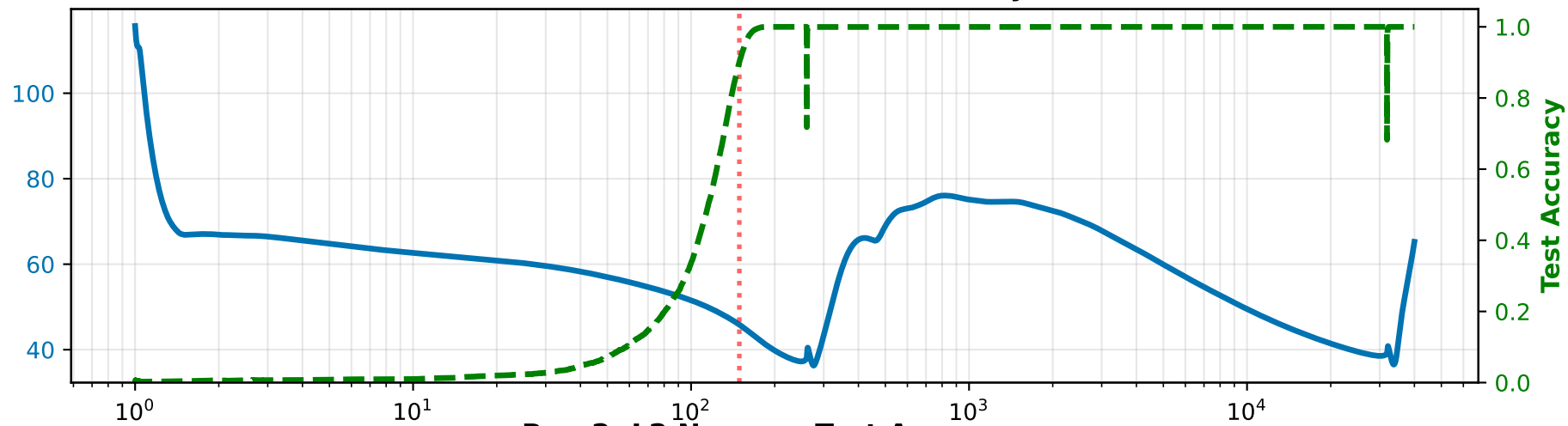


Dropout Gap (Post-Training Stress Test at rate=0.9)

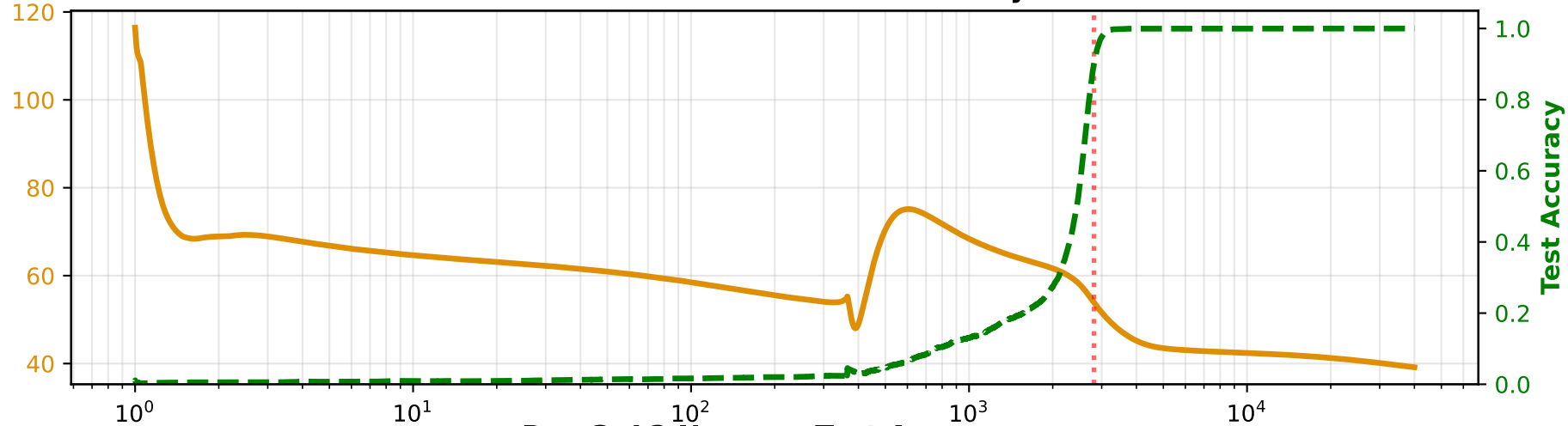


NOTE: Negative gap indicates dropout INCREASES accuracy (unexpected).
This suggests a bug in the dropout gap calculation or model has reached perfect accuracy.

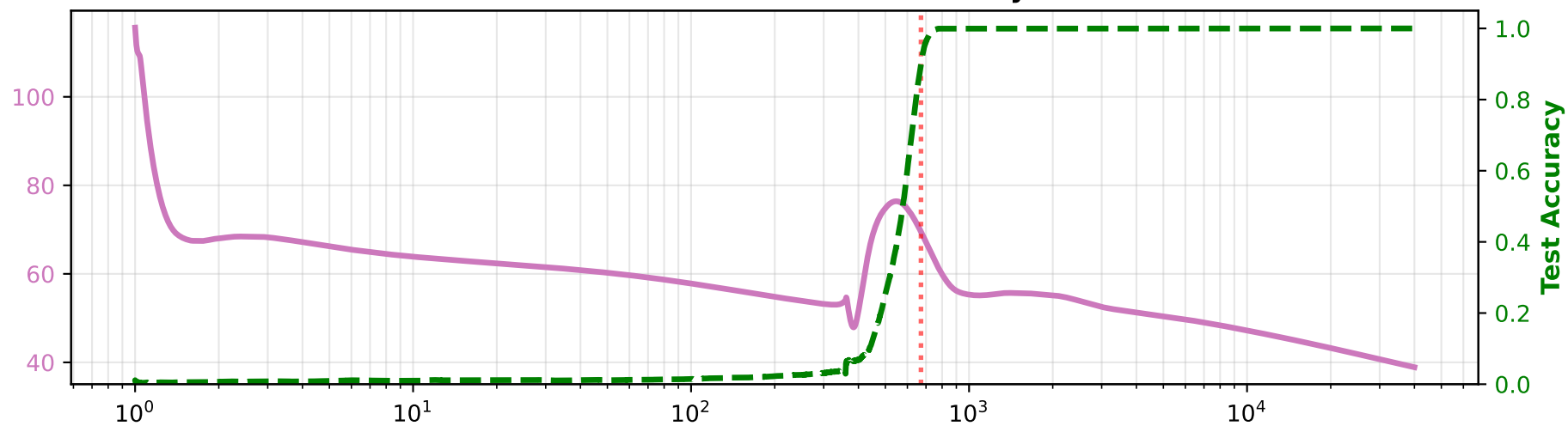
Run 1: L2 Norm vs Test Accuracy



Run 2: L2 Norm vs Test Accuracy



Run 3: L2 Norm vs Test Accuracy



Epoch (log scale)

KEY INSIGHTS AND CONCLUSIONS

1. STOCHASTIC GROKING BEHAVIOR:

- Grokking epochs vary dramatically: Run 1 at 149, Run 2 at 2,815, Run 3 at 671
- This is NOT consistent despite identical architecture and hyperparameters
- Suggests random seed strongly influences grokking timing
- Single-run predictions would be unreliable

2. L2 NORM VARIABILITY:

- Initial L2 norm: 115.7 to 116.4 (consistent)
- Final L2 norm: 39.2 to 65.2 (highly variable!)
- Total decay: 50.6 to 77.5 (different decay rates)
- L2 norm alone cannot predict grokking onset reliably

3. DROPOUT GAP SIGNAL (CONCERNING):

- All runs show NEGATIVE dropout gap (-0.97 to -0.98)
- This means dropout INCREASES accuracy after training
- Expected: dropout should DECREASE accuracy (stress test)
- This indicates either:
 - Bug in dropout gap calculation, OR
 - Model has reached perfect accuracy (ceiling effect)
- Dropout predictor is NOT currently usable

4. LOSS DECAY:

- All runs reach extremely low final loss (< 0.001)
- Indicates perfect or near-perfect training fit
- Loss curve does not clearly predict grokking

5. RECOMMENDATIONS:

- × L2 Norm: Not a reliable standalone predictor
- × Dropout Gap: Requires debugging before use
 - Need to fix dropout gap calculation
 - Consider next predictor in evaluation order (Spectral)
 - Ensemble multiple runs for any prediction attempt

6. EXPERIMENTAL RIGOR:

- Use minimum 3-5 runs before drawing conclusions
- Report mean, std, min, max of metrics
- Consider that grokking is inherently stochastic
- Average L2 norm behavior across seeds